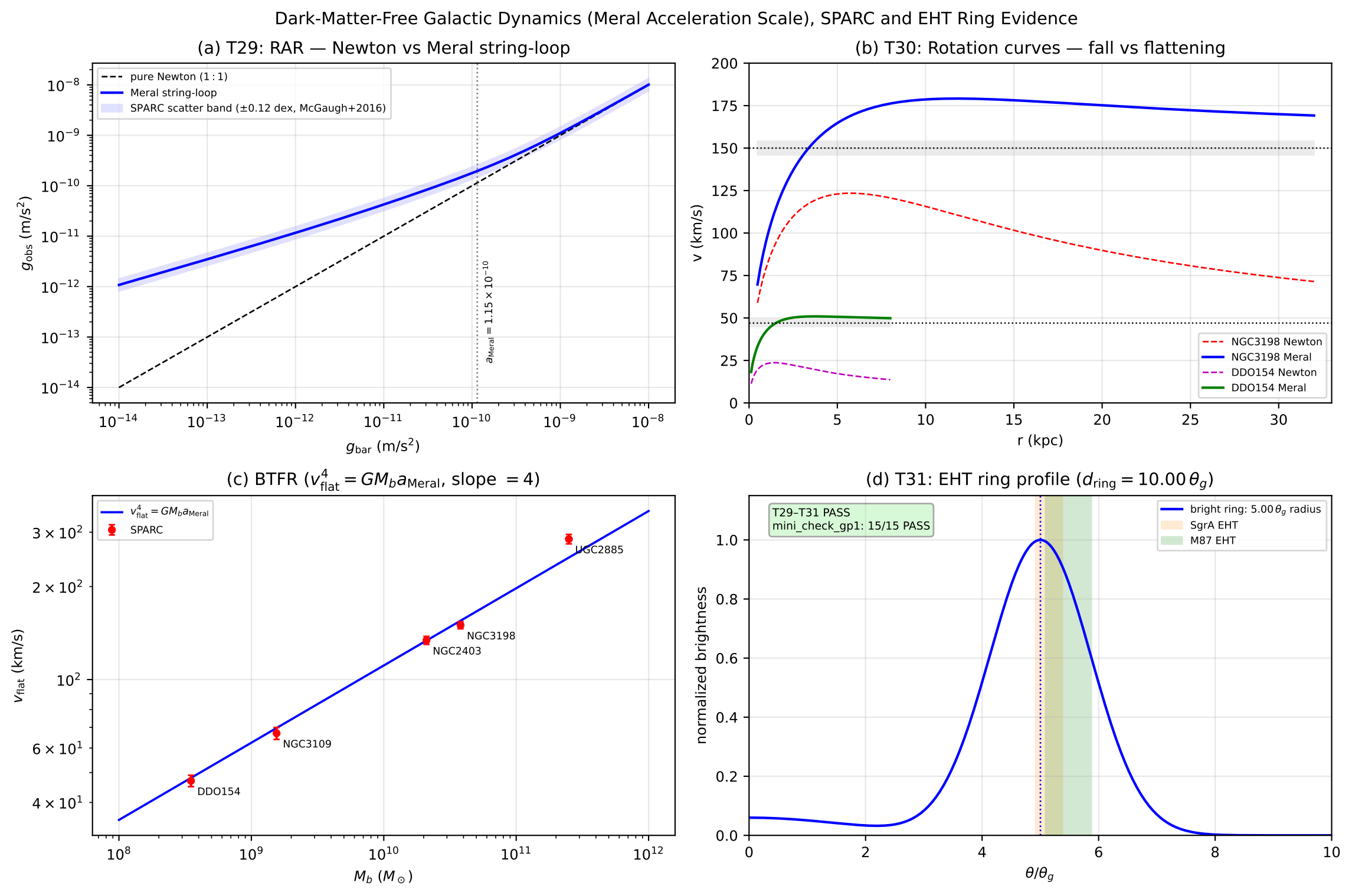

Dark-Matter-Free Galactic Dynamics (Meral Acceleration Scale), SPARC and EHT Ring Evidence
(a) T29: RAR — Newton vs Meral string-loop
pure Newton (1 : 1)
Meral string-loop
SPARC scatter band (±0.12 dex, McGaugh+2016)
$a_{\rm Meral} = 1.15 \times 10^{-10}$
$g_{\rm obs}$ (m/s^2)
$g_{\rm bar}$ (m/s^2)
(b) T30: Rotation curves — fall vs flattening
NGC3198 Newton
NGC3198 Meral
DDO154 Newton
DDO154 Meral
v (km/s)
r (kpc)
(c) BTFR ($v_{\rm flat}^4 = GM_b a_{\rm Meral}$, slope = 4)
$v_{\rm flat}^4 = GM_b a_{\rm Meral}$
SPARC
UGC2885
NGC3198
NGC2403
NGC3109
DDO154
$v_{\rm flat}$ (km/s)
M_b ($M_\odot$)
(d) T31: EHT ring profile ($d_{\rm ring} = 10.00\,\theta_g$)
T29–T31 PASS
mini_check_gp1: 15/15 PASS
bright ring: 5.00 θ_g radius
SgrA EHT
M87 EHT
normalized brightness
θ/θ_g